

CV Wave Theory: A Unified Model of Connected Speech Simplification

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Abstract

This paper introduces CV Wave Theory, a unified framework for understanding connected speech simplification. Human speech unfolds as a continuous stream shaped by the principle of articulatory economy. The theory formalizes spoken language as oscillations between consonantal closure (C, valleys) and vocalic openness (V, peaks), conceptualizing entire sentences as “one long word”. By modeling speech as a CV waveform, the model explains why phenomena such as linking, elision, reduction, and assimilation naturally occur: consonant clusters collapse into a single valley (e.g., next day → [neks deɪ]), adjacent vowels trigger linking or intrusion (e.g., law(r) and order), and non-prominent vowels reduce to schwa (e.g., going to → [gəʊnə]).

The framework extends beyond English. Cross-linguistic data from French (liaison), Spanish, Mandarin, and Russian show that syllable structures and vowel fusion also exhibit CV-wave dynamics, suggesting that CV Wave Theory may capture a universal principle of spoken fluency.

The contribution of this model is threefold:

1. **Integration** – it unifies multiple connected speech phenomena under a single explanatory framework.
2. **Visualization** – it provides a waveform-based representation of speech rhythm.
3. **Predictive power** – it anticipates which sounds are most likely to be weakened, elided, or linked.

CV Wave Theory bridges phonetics, phonology, and articulatory physiology, offering both a theoretical and pedagogical tool for analyzing natural speech flow.

Keywords: connected speech, CV structure, articulatory economy, phonology, speech rhythm, English pronunciation